

Project Initialization and Planning Phase

Date	4 July 20206
Team ID	x x x x x x
Project Title	Rising Water
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

Floods cause significant damage to lives, property, and the environment. Traditional forecasting methods often lack the accuracy and speed needed for timely disaster response. This project develops a **Machine Learning-based Flood Prediction System** that analyzes historical weather data to provide accurate flood risk predictions and support early warning and disaster management.

Project Overview	
Objective	Develop a Machine Learning-based Flood Prediction System that accurately predicts flood risk using historical weather data and provides timely predictions through a user-friendly web application.
Scope	The system analyzes weather parameters, predicts flood occurrence using machine learning models, and supports disaster management through early warning and decision-making.
Problem Statement	
Description	The project uses historical weather data to train machine learning models such as Decision Tree, Random Forest, KNN, and XGBoost. The best-performing model is integrated into a Flask web application to provide real-time flood predictions.

Impact	Solving these issues will result in improved operational efficiency, reduced risks, and an overall enhancement in the contribution to customer satisfaction and organizational performance.
Proposed Solution	
Approach	Employing machine learning techniques to enhance creditworthiness, creating a dynamic and adaptable system.
Key Features	Implementation of a machine learning-based credit model



	- Real-time decision-making for quicker loan approvals. - Continuous learning to adapt to evolving financial landscapes.
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Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	T4 GPU
Memory	RAM specifications	8 GB
Storage	Disk space for data, models, and logs	1 TB SSD
Software		

Frameworks	Python frameworks	Flask
Libraries	Additional libraries	scikit-learn, pandas, numpy, matplotlib, seaborn
Development Environment	IDE	Jupyter Notebook, pycharm
Data		
Data	Source, size, format	Kaggle dataset, 614, csv UCI dataset, 690, csv